Seshadri Naik Moode

PhD candidate in Civil Engineering (UPC 2026)

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Indian

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Summary

PhD researcher in Connected and Automated Vehicles (CAV) specialising in microscopic traffic simulation, safety analysis, and modelling of CAV systems. Experienced in devloping new models and analysing platooning strategies, collision risk, and vehicle behaviour under critical scenarios using Aimsun and Python. Research focuses on traffic flow dynamics and safety performance of CAVs in mixed traffic environments.

Academic degrees

01/2022 – Present
Barcelona, Spain

PhD candidate in Civil Engineering
Universitat Politècnica de Catalunya, BarcelonaTech
Member of the Barcelona Innovative Transportation (BIT) research group, concentrating on platooning connected autonomous vehicles (CAVs) with an emphasis on microscopic modeling and management strategies, under the guidance of Prof. Francesc Soriguera.

Academic stay: Research stay at KU Leuven Institute for Mobility, hosted by Prof. Claudio Roncoli (September 2025-January 2026)

09/2022 – Present

Urban Mobility
EIT Urban Mobility
Member of Doctoral Training Network funded by EIT Urban Mobility

2017 – 2019
Warangal, India

Transportation Engineering
National Institute of Technology, Warangal
Dissertation Title: Passenger Car Units for Urban Roads under Mixed Traffic Conditions. Under the supervision of Dr. Arpan Mehar, Assistant Professor, NIT Warangal.
Tools: PTV Vissim

2013 – 2017
RK Valley, India

Civil Engineering
Rajiv Gandhi University of Knowledge Technologies (RGUKT), RK Valley

Publications and Conference Proceedings

Zheng, X., Soriguera, F., Xie, C., & **Moode, S. N.** (2025, June). An Interaction-Aware Predictive Control Framework with Adaptive Gap Regulation in Connected and Automated Vehicle Platoon. In 2025 *IEEE Intelligent Vehicles Symposium (IV)* (pp. 1674-1680). IEEE.

Oriol, L., Martínez-Díaz, M., **Moode, S. N.**, Sala, M., & Soriguera, F. (2024). Platooning of connected automated vehicles on freeways: a microsimulation approach. *Transportmetrica B: Transport Dynamics*, 12(1), 2360543.

Moode, S. N., Soriguera, F., Zheng, X., & Martínez Díaz, M. (2024). Exploring safety in platoons of connected autonomous vehicles: investigating emergency braking conditions and collision severity. In Proceedings of the 12th Symposium of the European Association for Research in Transportation (hEART).

Moode, S. N., & Soriguera, F. (2023). Microscopic modeling of connected autonomous vehicles platooning: stability and safety analysis. *Transportation research procedia*, 71, 260-267.

Moode, S. N., & Soriguera, F. (2023). Platooning of connected autonomous vehicles in freeway traffic: state of the art. Transportation Research Procedia, 71, 268-275.

Conferences

06/2023 La Laguna, Tenerife	**Congreso de Ingenieria del Transporte (CIT 2023)** *Microscopic modeling of connected autonomous vehicles platooning: stability and safety analysis.* (*Oral Presentation*)
09/2023 Santander, Spain	**Euro Working Group on Transportation (EWGT)** *String-stable car-following model algorithm for platoons of connected autonomous vehicles.* (*Oral Presentation*)
06/2024 Espoo, Finland	**12th Symposium of the European Association for Research in Transportation. (HEART)** *Exploring safety in platoons of connected autonomous vehicles: investigating emergency braking conditions and collision severity.* (*Oral Presentation*)
09/2024 Heraklion, Greece	**Conference on Emerging Technologies (TRC-30)** *Definition of strategies to optimize the platooning of connected autornated vehicles on freeways.* (*Poster Presentation*)
09/2024 Heraklion, Greece	**The 5th Symposium on Management of Future Motorway and Urban Traffic Systems (MFTS)** *Examining Emergency Braking in Connected Autonomous Vehicle Platooning: Collision Risk and Severity Analysis.* (*Oral Presentation*)
01/2025 Washington, DC, USA	**104th Transportation Research Board Annual Meeting (TRB)** *Impact of lane management strategies of connected autonomous vehicle platoons on traffic flow: a simulation study.* (*Poster Presentation*)
06/2025 Zaragoza, Spain	**Congreso de Ingenieria del Transporte (CIT 2025)** *Lane Management Strategies for CAV Platoons: A Comparative Study of Mixed and Dedicated Lanes.* (*Oral Presentation*)
06/2025 Okinawa, Japan	**The 12th Triennial Symposium on Transportation Analysis conference (TRISTAN XII)** *Lane Management Strategies to Enhance Traffic Performance in Mixed Traffic Environments with Platoons of Connected Autonomous Vehicles* (*Oral Presentation*)

Skills

- **Simulation tools:** Aimsun, PTV
- **Programming & Data Analysis:** Python
- **Modelling:** Microscopic traffic modelling, CAV behaviour modelling
- **Domains:** Connected & Automated Vehicles, traffic flow theory, safety analysis

Awards

06/2023	**Young Researcher Award CIT 2023** *PTV Group Award for Young Researchers in Mobility and Accessibility in Transport* for an article on microscopic modeling of CAV platooning.
2022	**AGAUR PhD Grant** *Agència de Gestió d'Ajuts Universitaris i de Recerca. Genaralitat de Catalunya*

Languages

English C1	**Spanish** B1/B2	**Telugu**
Hindi	**Catalan** A2	

Doctoral Training Network Annual Forum

2022, 2023, 2024 Editions in Barcelona, Spain; Espoo, Finland; and Ghent, Belgium respectively.

Research Impact

- Research contributes to understanding **safety implications of CAV deployment in real-world traffic**
- Supports development of **methodologies for modelling and management of automated vehicle systems**
- Findings relevant for **traffic management strategies and future regulatory frameworks**

Research projects involved

09/2020 – 06/2021
Hyderabad, India

TIHAN IIT Hyderabad
Designed India's first autonomous vehicle testbed using AutoCAD, which set a national benchmark for research and development in the field of autonomous navigation.

- Led research on vehicle autonomy levels and traffic adaptation, providing critical insights that helped enhance the implementation of autonomous navigation in mixed Indian traffic conditions.
- Collaborated on safety system cvaluations and qualitative. studies, contributing to the development of reliable and context-aware navigation and collision warning systems, which improved overall safety for autonomous vehicles.

Teaching activities

Madanapalle, India

Assistant Professor
SVTM, Civil Engineering Department

- Used variely of learning modalities and support materials to facilitate learning process and accentuate presentations, including visual, aural and social learning modalities.
- Supported weekly transportation and fluid mechanics lab sessions, contributing to student engagement, comprehension and learning objectives.

Barcelona, Spain

Graduate Student Instructor
Universitat Politècnica de Catalunya
Assisted Prof. Francesc Soriguera in his teaching courses, particularly in Traffic and Operations courses.

Courses and training activities

12/2018
NIT Warangal, India
GIAN course on "Sustainable Mobility and Transportation: Innovative approaches and solutions"

08/2018
NII Warangal, India
GIAN course on "Transportation and the Environment"

05/2018
NIT Warangal, India
Participated in 5th Colloquium on Transportation systems engineering and Management

01/2018
NIT Warangal, India
GIAN course on "Decision Tools for Selection and Evaluation of Transportation Projects"

NIT Warangal, India
Participated in Five-day Faculty development program on Road safety issues and challenges

Internships

Hyderabad, India

M.Tech Internship
VR TECHNICHE Consultants Pvt.Ltd.
Preparation of project reports & schedules for award on long term fee collection for toll plazas.